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APPLICATIONS OF LASERS AND OTHER QUESTIONS OF QUANTUM ELECTRONICS

Study of Efficiency of Formation of Core–Satellite Gold- and Iron-Based Nanostructures by Laser Ablation Synthesis

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Abstract—We report the results of experimental work on laser synthesis of colloidal solutions of core–satellite nanostructures based on gold and iron (Au@Fe). The synthesis is optimized; the main result of optimization is an increase in the fraction of the synthesized Au@Fe nanoparticles (NPs) to 80% of the total number of NPs obtained while maintaining high synthesis productivity. The NPs synthesized using the optimized technology have a spherical shape with an average size of about 90 nm, and retain an optical extinction peak important for applications, which has a maximum of about 600 nm and a wide shoulder in the long-wavelength region. The Au@Fe nanoparticles are promising for use as a contrast agent in magnetic resonance and computed tomography, as well as sensitizers in photothermal and proton therapies.

Keywords: nanoparticles, satellite–core, magnetic particles, plasmon resonance, laser ablation in liquid, iron, gold

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1. INTRODUCTION

The continuing increase in the number of identified oncological diseases [1] requires constant modernization and expansion of methods for their diagnosis and therapy. Innovative approaches to nanotechnology can significantly expand the arsenal of both used and promising methods [2]. A topical issue in this context is the synthesis of new hybrid nanoparticles (NPs) consisting of several different materials and capable of simultaneously performing several functions, for example, participating in both imaging and therapy tasks. Optical methods play an important role in clinical practice due to their high spatial resolution and relative simplicity. Gold (Au)-based NPs are currently widely used in such methods [3–5]. The main limiting factor of optical methods—low penetration depth—can be overcome using physical fields freely propagating in biological tissues. One of the most striking examples here is a magnetic field and magnetic NPs that sensitize it, which are actively studied in many laboratories [6–8]. The most promising magnetic material is iron. Iron-based nanoparticles can be controlled by external magnetic fields in vitro and in vivo, which allows their use in a series of therapeutic and diagnostic modalities, including magnetic resonance imaging (MRI) [9], controlled drug delivery [10], and magnetic hyperthermia [11]. The combination of iron and gold within a single nanostructure appears very promising for the synthesis of new multifunctional NPs for biomedical applications.

Previous studies [12–14] have already demonstrated the synthesis of hybrid gold–iron NPs with various structures, including a core–satellite structure, which has shown high efficiency in imaging and therapy methods both in vitro and in vivo [15]. However, in most cases, these nanostructures were synthesized by colloidal chemistry methods that allow good control over the characteristics of the resulting NPs, but lead to surface contamination with precursor residues and by-products of chemical reactions, which can result in problems with secondary toxicity.

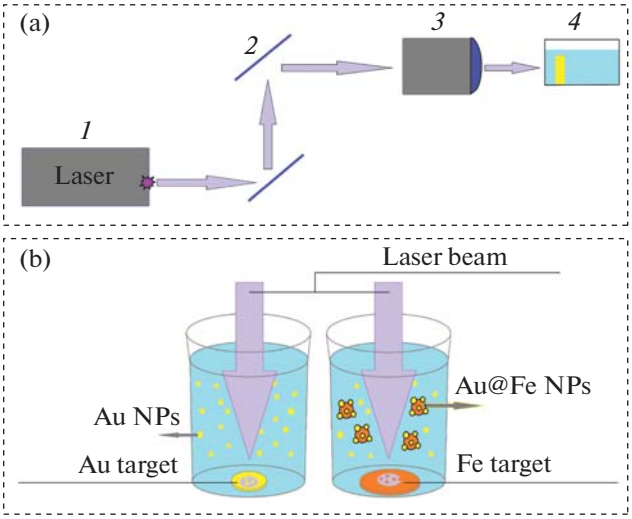

Fig. 1. (a) Scheme of the experimental setup: (*1*) femtosecond Yb:KGW laser, (*2*) mirrors, (*3*) galvanometric scanner, and (*4*) ablation chamber with a target. (b) Schematic diagram of the laser synthesis of Au@Fe NPs.

One of the promising methods for synthesizing NPs for biomedical applications is pulsed laser ablation in liquid [16, 17]. The key advantage of the method is the unique chemical properties of the surface of the resulting nanomaterials [18] and their colloidal stability without stabilizers. In addition, this method is very versatile and can be used to synthesize a wide range of both inorganic [19–22] and organic nanomaterials [23], and also makes it possible to produce NPs with various geometries [24, 25]. Pulsed laser ablation in liquid has already been successfully used to synthesize Au@Fe nanostructures with sizes suitable for biomedical applications, high optical absorption in the transparency window of biological tissues, and good magnetic properties [14, 26, 27]. The main disadvantage of using this method was a relatively low (about 5–10%) fraction of Au@Fe NPs in the total number of obtained nanoparticles.

This paper presents the results of optimizing the conditions of laser ablation synthesis, which make it possible to increase the fraction of Au@Fe NPs to 80% of the total number of obtained nanoparticles, and maintain high synthesis productivity and key dimensional and optical properties of the particles.

2. MATERIALS AND METHODS

The multistage technique used in this work is based on laser ablation of a Fe target in the presence of a pre-prepared colloidal solution of Au NPs. The scheme of the experimental setup and the synthesis process are shown in Fig. 1.

At the first stage, Au NPs were synthesized by laser ablation of a gold target placed in a cell with an aqueous NaCl solution with a concentration of 180 μM (2.98 mg per 50 mL). The following laser radiation parameters were used: pulse duration of 250 fs, wavelength of 1030 nm, pulse repetition rate of 100 kHz, and pulse energy of 30 μJ. For uniform ablation of the entire target surface, the laser beam was moved using a galvanometric scanner at a speed of 4 m/s along a spiral trajectory. After synthesis, the resulting colloidal solution was centrifuged (centripetal acceleration of 15000*g* and duration of 15 min). To remove relatively large NPs (more than 10 nm), all precipitated nanoparticles were removed, and NPs from the supernatant were used in further studies.

Next, laser synthesis of Fe NPs was performed in an ablation cell containing 25 mL of a colloidal solution of Au NPs (0.32 mg/mL), which subsequently became cores in Au@Fe nanostructures. The Fe target was ablated by varying the energy of laser pulses (5 μJ). The synthesis duration was from 3 to 30 min, and the mass of the ablated iron was measured gravimetrically (weighing the target before and after ablation).

At the last stage, the synthesized nanoparticles were magnetically separated to remove Au@Fe from the weakly magnetic fraction. To this end, a neodymium magnet was placed near the side wall of the vessel containing the colloidal solution, which led to the sedimentation of magnetic NPs near the magnet. After 10 min, all liquid was carefully removed from the vessel except for 300–500 μL near the magnet. The nanoparticles remaining in this volume were redispersed in 5 mL of liquid and used for further study. To calculate the synthesis productivity (the mass of synthesized Au@Fe NPs per unit time), the total mass

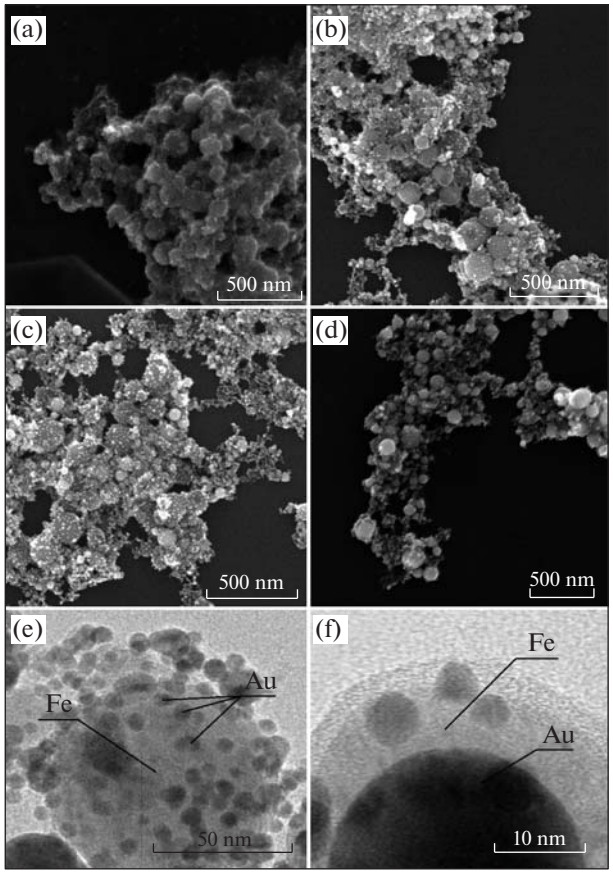

Fig. 2. Electron images of the synthesized nanostructures at different durations of Fe target ablation: (a) 3, (b) 7, (c) 13, and (d) 30 min, as well as high-resolution images for (e) individual Au@Fe NPs and (f) core–shell NPs with an Au core and an Fe oxide shell.

of the obtained NPs (estimated gravimetrically by drying 100–200 μL of the final colloid) was multiplied by the fraction of Au@Fe NPs in the total number of obtained nanoparticles and divided by the total time of laser synthesis (ablation of gold and iron targets).

The morphology, structure, and size of the synthesized nanoparticles were studied using a scanning electron microscope (SEM, MAIA 3, Tescan, Czech Republic) operating at a voltage of 0.1–30 kV. Samples for electron microscopy were prepared by applying 1 μL of the NP solution to a cleaned silicon substrate, followed by drying under normal conditions.

The number of core–satellite NPs and their fraction in the total number of obtained nanoparticles, as well as the construction of NP size distributions, were calculated by analyzing electron microscopy images using ImageJ software. To do this, the most representative images were selected and the sizes of all nanoparticles presented in these images were counted and measured (the total number of NPs for each sample was at least 500). The percentage content (yield) was obtained by counting the distinguishable Au@Fe structures in relation to the total number of nanoparticles.

Optical extinction spectra were measured using an MS 122 spectrophotometer (Sol Instruments, Republic of Belarus) in plastic optical cells with an optical path length of 10 mm.

3. RESULTS AND DISCUSSION

In this work, the main emphasis was placed on studying the effect of the Fe target ablation conditions on the fraction of Au@Fe nanostructures in the total number of synthesized nanoparticles. Preliminary, we found that the energy of laser pulses, their repetition rate, and focusing have little effect on the parameter in question. The most relevant parameter was the duration of the Fe target ablation process. As a result of the ablation synthesis, colloidal solutions of Au@Fe NPs were synthesized, differing in the duration of the Fe target ablation stage. Typical SEM images of the produced nanostructures are shown in Fig. 2.

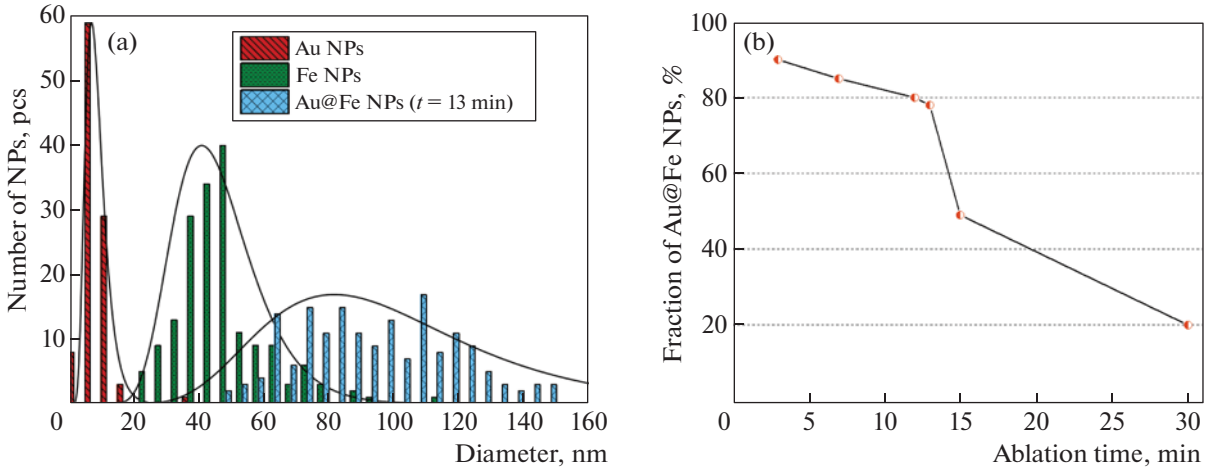

Fig. 3. (a) Size distributions of separately synthesized Au NPs and Fe NPs, as well as Au@Fe NPs, and (b) the dependence of the fraction of Au@Fe NPs in the total number of nanoparticles on the time of the Fe target ablation.

The obtained data indicate that the number of core–satellite NPs decreases with increasing Fe ablation time. Morphologically, Au@Fe NPs are relatively large spherical iron-based cores with an average size of about 75 nm, decorated with small spherical Au NPs with an average size of about 8 nm, as can be seen from Fig. 2 and the size distribution graphs shown in Fig. 3a. In our previous works [12, 15], we studied in detail the distribution of chemical elements in these nanostructures, and showed that their cores consist of iron oxides (mainly Fe_3O_4), with the satellites decorating them consisting of pure gold.

A detailed analysis of the dependence of the fraction of Au@Fe NPs in the total number of synthesized particles on the ablation time of the Fe target (Fig. 3b) shows that at relatively short synthesis durations (less than 13 min), the proportion of Au@Fe NPs remains high (more than 80%). At the same time, in the interval of durations between 13 and 30 min, there occurs a rapid decrease in the proportion of these nanostructures down to 20%. This effect is probably due to the fact that as Fe NPs are produced, the number of Au NPs available for the formation of Au@Fe nanostructures decreases. The optimal ablation time from the practical application point of view (high synthesis productivity while maintaining a high fraction of the required nanostructures in the final colloidal solution) is 13 min. This optimal duration depends on the initial concentration of Au NPs and the ablation productivity from the Fe target.

Note that the average size of the resulting Au@Fe NPs slightly exceeds the sum of the sizes of individual components of these nanostructures synthesized independently of each other by laser radiation with the same parameters. Indeed, Au@Fe NPs have a wide size distribution with an average value of about 90 nm (Fig. 3a), which is greater than the sum of the average size of separately synthesized Fe NPs and twice the average size of Au NPs. The observed effect can be explained by the influence of the magnetic separation stage. During this stage, only those nanoparticles were selected that managed to gather near the magnet for 10 min, which leads to the separation of a fraction of particles with a high magnetic moment, and so the smallest nanoparticles with an insufficient magnetic moment do not fall into it. Nevertheless, the final size of the obtained NPs is still in the range of sizes promising for biomedical applications.

The results obtained using spectrophotometry (Fig. 4) show a shift in the extinction peak of Au@Fe NPs to the long-wavelength region relative to the extinction peak of Au NPs and its significant broadening with an increase in the ablation time of the Fe target. The shift is in the range of 40–90 nm. In particular, the extinction peak shifts from 515 nm for Au NPs to 605 nm for Au@Fe NPs obtained by ablation of the Fe target for 30 min. The observed effect can theoretically be caused by a number of factors, including hybridization and coupling of plasmonic modes between Au NPs in aggregates [28], changes in the shape of Au NPs [29], and the formation of core–satellite or core–shell nanostructures. Note that the electron microscopy images show single core–shell structures (Fig. 2f), consisting of a gold core and an iron oxide shell, which probably also contribute to the observed optical extinction spectrum.

The presence of a broad optical extinction peak in Au@Fe NPs in the near-IR range is extremely important for biomedical applications of these nanoparticles, since this range contains the first window of relative transparency of biological tissues (650–950 nm), which allows NPs with high optical absorption to be efficiently used for methods such as photoacoustic imaging and photothermal therapy.

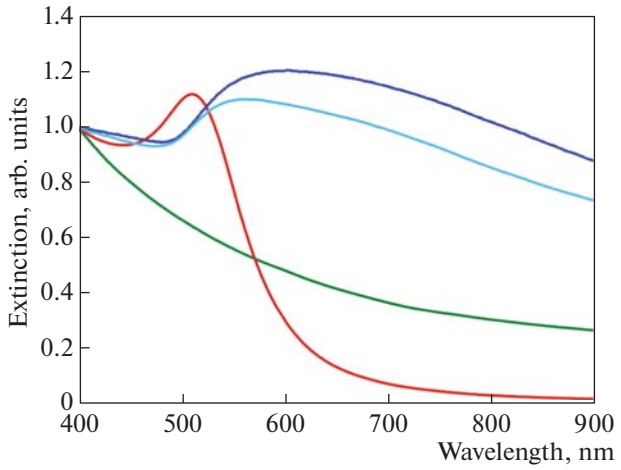

Fig. 4. Normalized optical extinction spectra of Au NPs (red curve), Fe NPs (green curve), and Au@Fe (blue and light blue curves for Fe ablation durations of 15 and 30 min, respectively).

4. CONCLUSIONS

In this work, we optimized the laser ablation synthesis of Au@Fe NPs, which made it possible to increase the yield of these structures to 80% of the total amount of synthesized nanoparticles, while maintaining high synthesis productivity (4 mg/h) and sizes acceptable for in vivo applications, as well as excellent optical properties. The combination of magnetic and plasmonic properties of Au@Fe NPs opens up prospects for the development of new methods for diagnosing and treating cancer and is a great advantage compared to single-component NPs. The achieved increase in the yield of these nanoparticles will improve the efficiency of their use in computer tomography, photoacoustic imaging and photothermal therapy due to an increase in the gold content in the NPs. In addition, the iron content in these nanoparticles makes them promising for use in magnetic hyperthermia and targeted magnetic delivery of drugs to a specified region.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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